DESIGN DOCUMENT

IFRS TAXONOMY & CURRENCY-AWARE NORMALIZATION

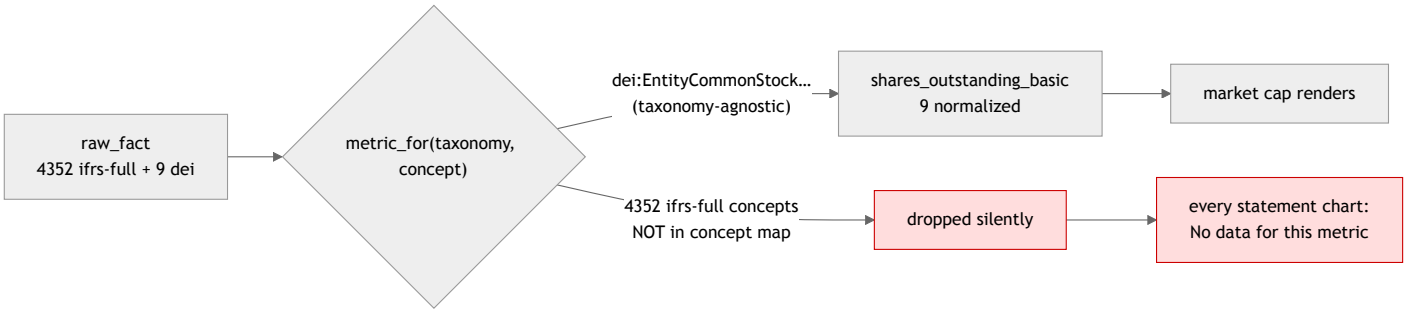
Status: Approved 2026-07-11 — recommended architecture adopted in full (incl. §5.2 debt policy and §6 currency scope). See §9. **Companions:** `docs/ifrs-currency-support-techspec.md` **Parent:** `docs/architecture.md` **Trigger:** INFY (Infosys Ltd, CIK 0001067491) renders a blank dashboard — 4361 raw facts ingested, only 9 normalized.

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1. Overview

Infosys files with the SEC as a **foreign private issuer**: its financial facts are tagged in the `ifrs-full` taxonomy, not `us-gaap`. Ingestion succeeds (748 filings, 4361 raw facts, 9 price points), but the normalization layer maps **only** `us-gaap` (51 curated concepts) plus a single `dei` concept. It contains **zero** `ifrs-full` mappings, so 4352 of 4353 IFRS facts are silently dropped. The one metric that survives — `shares_outstanding_basic` — comes through the taxonomy-agnostic `dei:EntityCommonStockSharesOutstanding` candidate, which is why market cap renders while every statement chart shows "No data for this metric."

Investigating the fix surfaced **three distinct gaps**. Only one is the blank screen; the other two are latent correctness risks that this document exists to separate, triage, and resolve durably.



The key structural finding: **the normalization engine is already taxonomy-agnostic.** `orchestrator.rs` calls `concept_map::metric_for(&f.taxonomy, &f.concept)` and normalizes whatever the table returns, with no `us-gaap` assumption anywhere on the path. IFRS produces nothing today for exactly one reason — the table has no IFRS rows. This is what makes Gap 1 additive rather than structural.

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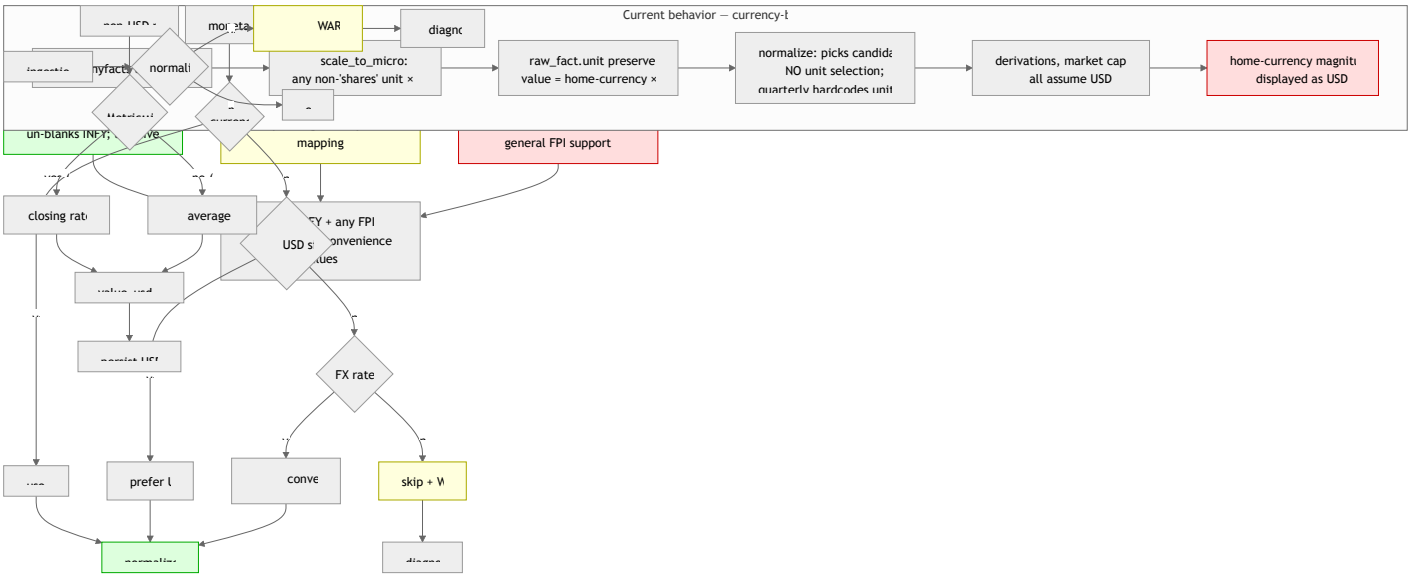
2. The three gaps

#	Gap	Root cause	Blocks INFY?	Correctness risk if ignored
1	**IFRS concept coverage** (incl. concept	No `ifrs-full` rows in `concept_map.rs`	**Yes — this is the blank screen**	—

#	Gap	Root cause	Blocks INFY?	Correctness risk if ignored
	drift across IFRS versions)			
2	**Debt semantics under IFRS 16**	IFRS capitalizes all leases as `LeaseLiabilities`; issuer carries no traditional borrowings	No	Mapping leases → debt inflates leverage and breaks cross-taxonomy comparability
3	**Currency-blind pipeline**	`scale_to_micro` and the normalize path treat every monetary unit as USD	No — INFY files USD convenience translations	**High** — any FPI without USD convenience values yields silently wrong numbers labelled USD

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3. Triage — what must be fixed, and when



- **Gap 1 — must fix now.** This is the reported bug. The fix is additive (add `ifrs-full` candidates to the existing per-metric arms) and unblocks INFY plus every IFRS filer that provides USD convenience translations.
- **Gap 2 — decide now, implement cheaply.** Not a code-architecture problem; a product/domain policy. It does not block the dashboard (INFY is debt-free). But the policy **must be decided before any IFRS debt concept is mapped**, or we bake in a silent cross-taxonomy inconsistency.
- **Gap 3 — guard now, full fix before general FPI support.** INFY is saved only by luck: Infosys tags USD convenience values in its 20-F, which the pipeline scales correctly ($3.162B `ProfitLoss` FY25, $10.94B `Revenue` FY18 both verified accurate). The moment an issuer reports only in its home currency, today's pipeline stores home-currency magnitudes labelled USD — silently wrong. We add a **loud diagnostic now** (so this never manifests as a blank/garbage screen again) and schedule the FX-conversion fix as its own work package.

Do not fix: nothing here is a no-op. All three warrant action; they differ in urgency and altitude.

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4. Gap 1 — IFRS concept coverage

4.1 What the data actually supports

I matched every internal metric against the `ifrs-full` concepts Infosys actually reports (counts are fact rows in the production DB). **14 of ~16 statement metrics have a clean, directly-mappable IFRS concept:**

Internal metric	IFRS concept (fact count)	Note
Revenue	`RevenueFromContractsWithCustomers` (21), `Revenue` (3)	**concept drift** — modern IFRS 15 tag vs. legacy; needs both candidates
CostOfRevenue	`CostOfSales` (24)	clean
GrossProfit	`GrossProfit` (24)	clean
OperatingIncome	`ProfitLossFromOperatingActivities` (24)	clean — issuer *does* tag it
NetIncome	`ProfitLoss` (24)	clean
EpsBasic / EpsDiluted	`Basic/DilutedEarningsLossPerShare` (24)	clean, USD/shares
SharesOutstandingBasic	`WeightedAverageShares` (24)	clean
SharesOutstandingDiluted	`AdjustedWeightedAverageShares` (24)	naming: "adjusted" = diluted
CashAndEquivalents	`CashAndCashEquivalents` (28)	clean
TotalAssets / TotalLiabilities	`Assets` (16) / `Liabilities` (16)	clean
TotalEquity	`EquityAttributableToOwnersOfParent` (16), `Equity` (32)	parent-first, incl-NCI fallback (mirrors US-GAAP ordering)
CashFromOperations	`CashFlowsFromUsedInOperatingActivities` (24)	clean
CapitalExpenditures	`PurchaseOfPropertyPlantAndEquipmentClassifiedAsInvestingActivities` (9)	sign already normalized by `apply_sign`
DepreciationAmortization	`DepreciationAndAmortisationExpense` (24)	clean
PropertyPlantAndEquipmentNet	`PropertyPlantAndEquipment` (32)	clean

The read-time derivations need **no change**: free cash flow (`ProfitLoss` + D&A – capex), the capex PP&E roll-forward fallback (`PropertyPlantAndEquipment` + D&A), and operating margin all resolve from the metrics above once the

concepts are mapped.

4.2 Options

Option A — inline `ifrs-full` candidates in the existing arms. *(Recommended.)* Extend each `concepts_for` arm with its IFRS candidates after the `us-gaap` ones. Keeps a single source of truth per metric; the ordered-candidate mechanism already handles multi-taxonomy fallback; `metric_for` reverse lookup already iterates all candidates. ~20 concept lines.

Option B — externalize the map to a data file (TOML/CSV) loaded at startup. More scalable for many taxonomies, but introduces a load path, a parse-error surface, and drift between file and `Metric` enum. Not justified at two taxonomies.

Concept drift within a taxonomy (e.g. IFRS 15 changed `Revenue` → `RevenueFromContractsWithCustomers`) is handled by the **same ordered-candidate list** we already use for US-GAAP synonyms — no new mechanism.

4.3 Recommendation

Option A, plus two **forward-looking guards** (§7) so a future metric can't silently regress IFRS coverage and so a coverage collapse never again surfaces as a blank screen.

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5. Gap 2 — debt semantics under IFRS 16

5.1 The problem

Under **IFRS 16** every lease is capitalized as a `LeaseLiabilities` line. Infosys carries essentially no financial borrowings (`Borrowings`: 1 fact) — its balance-sheet "debt" is almost entirely lease liabilities (22 facts). Under **US-GAAP (ASC 842)** operating-lease liabilities are also on-balance-sheet, but our `us-gaap` map deliberately routes them to **neither** `LongTermDebt` **nor** `CurrentDebt` — only interest-bearing borrowings count as debt.

If we map IFRS `LeaseLiabilities` → debt, an IFRS filer's leverage would include operating leases that an otherwise-identical US filer's would exclude. That is a **cross-taxonomy comparability defect**: the same metric would mean two different economic things depending on where the company files.

5.2 The governing principle & options

> **Economic-equivalence rule:** a metric means the same economic thing regardless of taxonomy. An IFRS concept may be mapped to a metric only if it is economically equivalent to that metric's US-GAAP concept — not merely "close."

Option A — map interest-bearing borrowings only; exclude leases. *(Recommended.)* `LongTermDebt` ← `NoncurrentBorrowings` / `Borrowings`; `CurrentDebt` ← `CurrentBorrowings`. Leases are excluded, exactly as for US-GAAP. Preserves comparability. For Infosys this yields ~zero debt — which is *correct*.

Option B — count leases as debt on both sides. Consistent across taxonomies, but changes existing US-GAAP numbers (larger blast radius), and conflicts with how most screeners report "debt." Rejected as default.

Option C — introduce a distinct `LeaseLiabilities` metric. Cleanest long-term: surface leases explicitly rather than forcing them into "debt." Deferred — adds a metric and UI surface; not required to render INFY correctly.

5.3 Recommendation

Adopt the economic-equivalence rule as written policy. Implement **Option A** now. Keep **Option C** on the roadmap for when lease disclosure becomes a product requirement.

Approved (2026-07-11). Lease liabilities are **not** debt: only interest-bearing borrowings map to `LongTermDebt` / `CurrentDebt`, matching the US-GAAP treatment. For Infosys this correctly yields ~zero debt. A separate `LeaseLiabilities` metric (Option C) is deferred until lease disclosure is a product requirement.

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6. Gap 3 — currency-aware normalization (the architectural core)

6.1 Why today's pipeline is currency-blind

Three concrete facts from the code:

- `scale_to_micro` (`companyfacts.rs:118`) multiplies **any** unit except `shares` by 10^6 — it never inspects the currency.
- The normalize path (`orchestrator.rs`) selects candidates and best-per-period **without ever filtering on unit**; the quarterly branch hardcodes `unit: "USD"` (line 185); the annual/instant branches preserve `f.unit` but everything downstream assumes USD.
- `normalized_fact` already has `fx_rate_micro`, `fx_rate_source`, `fx_rate_date`, `original_value`, `original_unit` columns and there is an `fx_rate` table — **the FX scaffolding exists but is wired to `None` everywhere.**

INFY escapes this only because Infosys files USD convenience translations (4032 of its facts are USD). A pure-INR filer would flow home-currency magnitudes straight through as "USD."

The `require_usd` guard on the **market-data** adapter (`market_data.rs:106`) already embodies the correct principle for quotes — *reject rather than mislabel*. Statement facts have no equivalent guard. The fix extends that principle to the fact pipeline.

6.2 Options

Option A — USD-preference candidate selection. When a concept is tagged in multiple currencies for a period, prefer the USD-unit fact and drop non-USD when a USD sibling exists. Cheap; fully fixes convenience-translation filers (INFY) and prevents accidental home-currency pickup. **Does not** help pure-foreign-currency filers.

Option B — FX conversion at normalization. *(The durable fix.)* For a monetary fact whose currency ≠ USD, convert to USD **using a rate matched to the fact's nature** (§6.5), store the converted value, and record provenance in the **existing** columns (`original_value` , `original_unit` , `fx_rate_micro` , `fx_rate_source` , `fx_rate_date`). Requires an `FxRateSource` and populating the `fx_rate` table.

Option C — reject + diagnostic. If a monetary fact is non-USD and cannot be converted, **skip it and emit a WARN ingestion event** rather than store a mislabelled number. This is the accuracy-preserving floor and aligns with the project's hard-accuracy rule.

6.3 Recommended architecture — a layered currency-resolution stage

Introduce an explicit **currency-resolution step** between raw-fact selection and normalized-fact persistence, applying a strict ladder per monetary fact:

Supporting structural changes:

1. **Treat currency as first-class.** `raw_fact.unit` already carries it; stop assuming USD downstream. The quarterly branch must stop hardcoding `unit: "USD"` and instead flow the resolved currency.
2. **Split `scale_to_micro` 's two responsibilities** — "is this monetary?" (scale ×10^6) vs. "which currency is it?" (drives resolution). Scaling stays; currency detection becomes explicit.
3. **`FxRateSource` trait** feeding the `fx_rate` table (a daily-rate cache), mirroring the existing `MarketDataSource` adapter shape. Rate keyed by (currency, date); the rate *selection* is by fact nature (§6.5), not a single fixed date.
4. **Provenance is mandatory** whenever a value is FX-converted: the converted USD value is the fact; `original_value` / `original_unit` / `fx_rate_*` record how we got there, upholding the product's auditability principle.

6.5 Rate selection — match the rate to the fact (IAS 21 / ASC 830)

A single "period-end rate for everything" is **incorrect** for the income statement: applying one day's rate to a full year of revenue misstates it. Presentation-currency translation matches the rate to the nature of the fact, and the system already classifies this via `Metric::is_instant()` :

Fact nature	Metrics	Rate
Stock (instant / balance sheet)	cash, debt, assets, liabilities, equity, PP&E	**Closing spot** rate at the period-end date
Flow (duration / income & cash-flow)	revenue, net income, operating income, D&A, capex, operating cash flow	**Average** rate over `[period_start, period_end]`

The `fx_rate` table stays a **pure daily-rate cache** (no schema change): the closing lookup reads one row at/near period-end; the average is *derived* by reading all daily rows in the period and averaging. Only the tracked metrics × their periods need rates, so lookups per company are bounded.

Consistency rule. A derived metric (e.g. free cash flow = net income + D&A – capex) must not mix company-provided USD convenience values with market-rate conversions — they carry different implied rates and the sum is incoherent. Keep a single basis per statement: prefer USD convenience values uniformly when they exist (WP3a); only market-convert when they don't, applying one methodology across all flow facts in the derivation.

6.4 Recommendation

Ship **Option A + Option C** (they are cheap, and together they make the pipeline *honest* — correct for convenience-translation filers, loud-and-safe for everyone else) **and Option B** (FX conversion), which unlocks general FPI support. The FX-source choice is in the tech spec.

Approved (2026-07-11, revised). All three gaps are fixed in a **single phase** — there are no intermediate ship points. The layered resolution ladder (prefer USD → FX-convert with recorded provenance → skip + diagnostic) is the target architecture, with rate selection by fact nature (§6.5). WP1, WP2, WP3a, WP3b, and the guards (WP-G) all land together; the WP3a-before-WP3b relationship is an internal build order within the phase, not a separate release.

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7. Forward-looking guards — stop silent recurrence

The deeper defect behind all three gaps is **silence**: 4352 dropped facts produced a blank screen with no error. These guards convert each failure mode into a loud, testable signal.

Guard	Prevents	Mechanism
Coverage-matrix test	Gap 1 regressions	Unit test over `Metric::ALL`: any metric with `us-gaap` candidates must also declare `ifrs-full` candidates (or be explicitly annotated US-only, e.g. bank concepts). Fails CI when a new metric forgets IFRS.
Raw-to-normalized ratio diagnostic	Gaps 1 & 3 in the wild	At end of ingestion, if `normalized_facts / raw_facts` falls below a threshold (e.g. the 9/4361 seen here), emit a **WARN** ingestion event naming the dominant unmapped taxonomy. The blank screen becomes a visible diagnostic.
Economic-equivalence review checklist	Gap 2	Documented rule (§5.2) + a checklist item on any concept-map change: "is this IFRS concept economically equal to the US-GAAP concept for this metric?"
Currency provenance assertion	Gap 3	Test: no `normalized_fact` may carry a non-USD `unit`; any FX-converted fact must have non-null `original_unit` + `fx_rate_*`.

8. Decision summary

Gap	Verdict	Chosen approach	Status
1 — IFRS coverage	**Fix now**	Inline `ifrs-full` candidates (Opt A) + coverage-matrix test	**Approved**
2 — Debt semantics	**Decide now, implement cheaply**	Economic-equivalence rule; borrowings-only, exclude leases (Opt A); `LeaseLiabilities` metric deferred (Opt C)	**Approved — leases are not debt**
3 — Currency	**Full fix this cycle**	USD-preference + reject/diagnostic (Opts A+C) + FX conversion via `FxRateSource` (Opt B), rate selected by fact nature (§6.5)	**Approved — WP3a + WP3b both this cycle**
Cross-cutting	**Fix now**	Four guards in §7	**Approved**

All four rows ship in one phase — no intermediate milestones. The blank screen closes with Gap 1 alone, but Gaps 1–3 and the guards are delivered together: Gap 1 (IFRS coverage), Gap 2 (leases are not debt), and Gap 3 end-to-end (USD-preference + diagnostic + FX conversion, rate selected by fact nature, §6.5) all land in the same release. The WP1→WP2→WP3a→WP3b lettering denotes internal build order within the phase, not separate deliverables.

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9. Approvals

Date	Decision	Approved by
2026-07-11	Recommended architecture adopted **in full** — no deviations.	Product owner
2026-07-11	**Gap 2 (debt):** lease liabilities are **not** debt; map interest-bearing borrowings only, for parity with the US-GAAP treatment. `LeaseLiabilities` metric deferred.	Product owner
2026-07-11	**Gap 3 (currency):** layered resolution ladder is the target architecture. WP3a (USD-preference + reject/diagnostic) ships with WP1.	Product owner
2026-07-11 (revised)	**Gap 3 (currency), scope change: WP3b (FX conversion) moved into this cycle.** Rate selection is by fact nature — closing spot rate for balance-sheet (stock) facts, period-average rate for income/cash-flow (flow) facts (§6.5), correcting the earlier "period-end for everything" simplification.	Product owner
2026-07-11	**Guards (§7):** all four adopted and ship with WP1.	Product owner

Date	Decision	Approved by
2026-07-11 (revised)	**Single-phase delivery:** all gaps (WP1, WP2, WP3a, WP3b) and the guards (WP-G) are implemented and shipped in **one phase** — no intermediate milestones. WP letters denote internal build order only.	Product owner

Implementation status: not started. Deferred at the product owner's request while a separate agent is editing source files, to avoid write conflicts. This document and its companion tech spec are the authoritative record until implementation begins.